

Worksheet 1 — Designing Investigations

2025-08-26

1 Purpose

Practise planning a valid, reliable investigation using SI units and scientific notation.

2 Outcomes (map)

- INS11-12-2: Designs and evaluates investigations
- INS11-12-3: Conducts investigations to collect valid and reliable data
- INS11-12-7: Communicates scientific understanding

3 Tasks

- 1) Identify the independent, dependent and controlled variables for the scenario below. Specify ranges in SI units.
- 2) Sketch a method that addresses reliability, validity and safety.
- 3) Identify potential sources of error and how to minimise them.

TO-VERIFY: Insert context aligned with current unit scope.

4 Evidence

Attach a photo of your setup (if practical) or a diagram with labels and units.